

B.Tech IV Year I Semester (R20) Supplementary Examinations April/May 2025

BLOCKCHAIN TECHNOLOGY AND APPLICATIONS

(Common to IT, AI&DS, CSE (AI), CSE (AI&ML), CSE (DS), CSE (IoT), CSE, CSE (CS), CS&D and AI&ML)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|---|----|
| (a) What is the primary opportunity of using blockchain in various industries? | 2M |
| (b) What is Bitcoin? | 2M |
| (c) What is the primary purpose of hashing in blockchain technology? | 2M |
| (d) Define Merkle Tree and its role in blockchain. | 2M |
| (e) What are the key architecture considerations for designing blockchain applications? | 2M |
| (f) What is a cryptographic token, and how is it used in blockchain solutions? | 2M |
| (g) What is MetaMask, and how is it used to interact with decentralized applications? | 2M |
| (h) What is My Ether Wallet, and how is it used to interact with the Ethereum network? | 2M |
| (i) What is Hyperledger Fabric, and how does it enable private and secure blockchain networks? | 2M |
| (j) What is the InterPlanetary File System (IPFS), and how does it relate to blockchain technology? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

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|-----------|--|-----|
| 2 | (a) Explain the concept of decentralized applications and their role in the blockchain ecosystem. | 5M |
| | (b) Describe the key characteristics of blockchain technology. | 5M |
| OR | | |
| 3 | Compare and contrast public blockchain environments with other types of blockchain networks. | 10M |
| 4 | (a) Discuss the consensus mechanisms in blockchain, including proof-of-work (PoW) and proof-of-stake (PoS). | 5M |
| | (b) Describe the different types of blockchain nodes in a peer-to-peer network, including full nodes, light nodes, and mining nodes. | 5M |
| OR | | |
| 5 | (a) Discuss the security measures implemented in blockchain technology to protect data and transactions. | 5M |
| | (b) Explain the concept of public-key cryptography and its role in securing blockchain networks. | 5M |
| 6 | Discuss the importance of a reference architecture in designing blockchain solutions and explain how it can be applied to various use cases. | 10M |
| OR | | |
| 7 | Explain the key architecture considerations for designing blockchain applications, including scalability, security, and interoperability. | 10M |
| 8 | Discuss the Ethereum development tool stack, including Truffle, Ganache, and Open Zeppelin contracts. Explain their roles and interactions. | 10M |
| OR | | |
| 9 | Explain the concept of unit testing in smart contract programming and its importance in ensuring contract correctness and security. | 10M |
| 10 | (a) Describe the FabCar use case implementation in Hyperledger Fabric. | 5M |
| | (b) How do client applications invoke Chaincode functions in Hyperledger Fabric? | 5M |
| OR | | |
| 11 | Explain the relationship between quantum computing and blockchain technology, including potential impacts and benefits. | 10M |

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

BLOCKCHAIN TECHNOLOGY AND APPLICATIONS

(Common to IT, AI&DS, CS (AI), CSE (AI&ML), CSE (DS), CSE (IOT), CSE, CSE (CS), CS&D and AI&ML)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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|--|----|
| (a) Differentiate between Centralized and Decentralized Applications (Valid any two difference). | 2M |
| (b) Need for Miners in Block Chain. | 2M |
| (c) What is the difference between a token and a crypto currency in blockchain? | 2M |
| (d) What is consensus in blockchain, and why is it essential? | 2M |
| (e) Describe the key components of the Blockchain Solutions Reference Architecture. | 2M |
| (f) What are the different types of Blockchain applications? | 2M |
| (g) What is Infura, and how is it used to access the Ethereum network? | 2M |
| (h) What is the difference between an Ethereum account and a smart contract account? | 2M |
| (i) What is Chain code, and how is it used in Hyper ledger Fabric? | 2M |
| (j) How do client applications invoke Chain code functions in Hyper ledger Fabric? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

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|---|----|
| 2 (a) What are the major challenges articulated in the adoption of blockchain technology? | 5M |
| (b) Discuss the evolution of computer applications from centralized to decentralized systems. | 5M |
- OR**
- | | |
|---|----|
| 3 (a) What is a fork in the blockchain, and how does it impact the network? | 5M |
| (b) Describe the different types of players in the blockchain ecosystem. | 5M |
- | | |
|---|----|
| 4 (a) Explain the concept of hashing and its significance in blockchain technology. | 5M |
| (b) Describe how it ensures data integrity and security. | 5M |
- OR**
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|---|-----|
| 5 Describe the Merkle Tree data structure and its role in blockchain. How does it facilitate efficient data verification and storage? | 10M |
|---|-----|
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|---|-----|
| 6 Explain the concept of cryptographic tokens and their role in blockchain solutions. Discuss their advantages and disadvantages. | 10M |
|---|-----|
- OR**
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|--|-----|
| 7 Explain the Blockchain Relevance Evaluation Framework and its significance in architecting blockchain solutions. | 10M |
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| 8 Explain the use of Infura and Ether Scan in interacting with the Ethereum network and analyzing smart contract behavior. | 10M |
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- OR**
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|---|-----|
| 9 Design and implement a simple decentralized application (dApp) using Ethereum smart. Contracts and the Truffle framework. Explain the code and its functionality. | 10M |
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|---|-----|
| 10 Discuss the concept of Zero-Knowledge Proofs, including their application in blockchain and benefit. | 10M |
|---|-----|
- OR**
- | | |
|---|----|
| 11 (a) Describe the FabCar use case implementation in Hyper ledger Fabric. | 5M |
| (b) Describe the concept of Self-Sovereign Identity and its significance in blockchain. | 5M |

B.Tech IV Year I Semester (R20) Regular Examinations December/January 2024

BLOCKCHAIN TECHNOLOGY AND APPLICATIONS

(Common to IT, AI&DS, CSE (AI), CSE (AI&ML), CSE (DS), CSE (IOT) and CSE)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

- | | |
|--|----|
| (a) What is Blockchain? | 2M |
| (b) What are the different areas of blockchain technology? | 2M |
| (c) What is consensus? | 2M |
| (d) What is a currency token? | 2M |
| (e) What are the different obstacles for use of block chain? | 2M |
| (f) How cryptographic tokens work? | 2M |
| (g) What is Ethereum? | 2M |
| (h) What is truffle? | 2M |
| (i) What is Hyper ledger fabric? | 2M |
| (j) What is an Initial Coin Offering (ICO)? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- | | |
|---|----|
| 2 (a) Explain decentralized applications of blockchain in detail. | 5M |
| (b) Explain stages in blockchain evolution in brief. | 5M |
| OR | |
| 3 (a) Elaborate types of players in blockchain ecosystem. | 5M |
| (b) Discuss features of consortium blockchain. | 5M |
| 4 (a) Elaborate structure of a block. | 5M |
| (b) Elaborate life cycle of blockchain transaction. | 5M |
| OR | |
| 5 (a) Explain different types of blockchain nodes. | 5M |
| (b) How to use blockchain for data storage? | 5M |
| 6 (a) Explain hybrid blockchain with its advantages and disadvantages. | 5M |
| (b) Elaborate blockchain relevance evaluation framework. | 5M |
| OR | |
| 7 (a) Discuss obstacle for use of block chain. | 5M |
| (b) Discuss four applications of blockchain. | 5M |
| 8 (a) Explain tuna fish tracking use case in brief. | 5M |
| (b) Elaborate Ethereum Virtual Machine in detail. | 5M |
| OR | |
| 9 (a) Explain my ether wallet in brief. | 5M |
| (b) What are the different applications of smart contract? | 5M |
| 10 (a) Explain in detail about hyperledger fabric transaction flow. | 5M |
| (b) Explain Inter planetary File System in brief. | 5M |
| OR | |
| 11 (a) Elaborate FabCar use case implementation in brief. | 5M |
| (b) How does IOT work with blockchain? Explain benefits of blockchain in IOT. | 5M |

B.Tech IV Year I Semester (R20) Advanced Supplementary for MOOCs (Conventional mode) Examinations July 2024

BLOCKCHAIN TECHNOLOGY AND APPLICATIONS
(For 2020 Regular & 2021 Lateral Entry admitted batches only)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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|---|----|
| (a) Write the Advantages of Blockchain Technology. | 2M |
| (b) Differentiate the Private blockchain from public blockchain. | 2M |
| (c) What is transaction? Define in context to blockchain technology. | 2M |
| (d) What is a Bitcoin wallet? | 2M |
| (e) Define the various types of Ethereum Nodes. | 2M |
| (f) What is the role of Ethereum in the blockchain? | 2M |
| (g) What is a variable in solidity? | 2M |
| (h) Outline the difference between struct and Array in Solidity. | 2M |
| (i) What do you understand by Sybil attacks in blockchain technology? | 2M |
| (j) Write the short notes on Distributed Ledger. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

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|--|----|
| 2 (a) How is blockchain distributed database different from traditional databases? | 5M |
| (b) What is private blockchain? Give examples. | 5M |
- OR**
- | | |
|---|----|
| 3 (a) Explain the core components of blockchain architecture? | 5M |
| (b) How do you ensure the confidentiality, integrity, and availability of data in a blockchain network? | 5M |
- 4 What is tokenization? Write the importance of tokenization in blockchain Technology. 10M
- OR**
- 5 Write the short notes on: 10M
(i) Cryptography, (ii) Immutability, (iii) Transparency.
- 6 What is the Consensus mechanism in Blockchain? Discuss any two Consensus Algorithm. 10M
- OR**
- 7 Differentiate Permission less Models with Permissioned Models with help of suitable example. 10M
- | | |
|---|----|
| 8 (a) Why is Ethereum an important development in the blockchain ecosystem? | 5M |
| (b) Explain the Hyper ledger reference architecture. | 5M |
- OR**
- 9 Illustrate the function of smart contract. How you Deploy smart contracts on a blockchain? 10M
- 10 Discuss the various kind of challenges occurred during the implementation of any blockchain application. 10M
- OR**
- | | |
|--|----|
| 11 (a) What is Crypto currency? How does blockchain support Bitcoin? | 5M |
| (b) Write the function of Bitcoin wallet. How can you choose a Bitcoin wallet? | 5M |
